Whole-body Motion Planning and Control for underactuated wheeled-bipedal robots

FRANCESCA DI SANTO - 1998148
DARIO FINOCCHIARO - 1988366
ARIANNA NAVARRA - 2195207

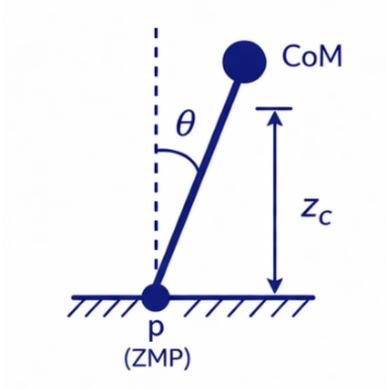
Problem

1. **Unstable underactuated system:** the robot has only two ground contact points, making dynamic balance challenging

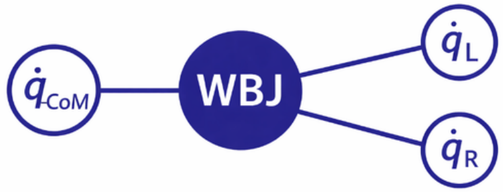
2. **Highly coupled dynamics:** balance, movement of wheels and locomotion of legs are strongly interconnected

3. **Limitations of traditional control method:** traditional control methods are unable to solve the problem of limited exercise

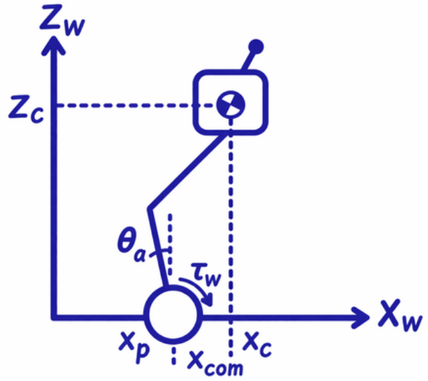
Solution



WIPM-based modelling

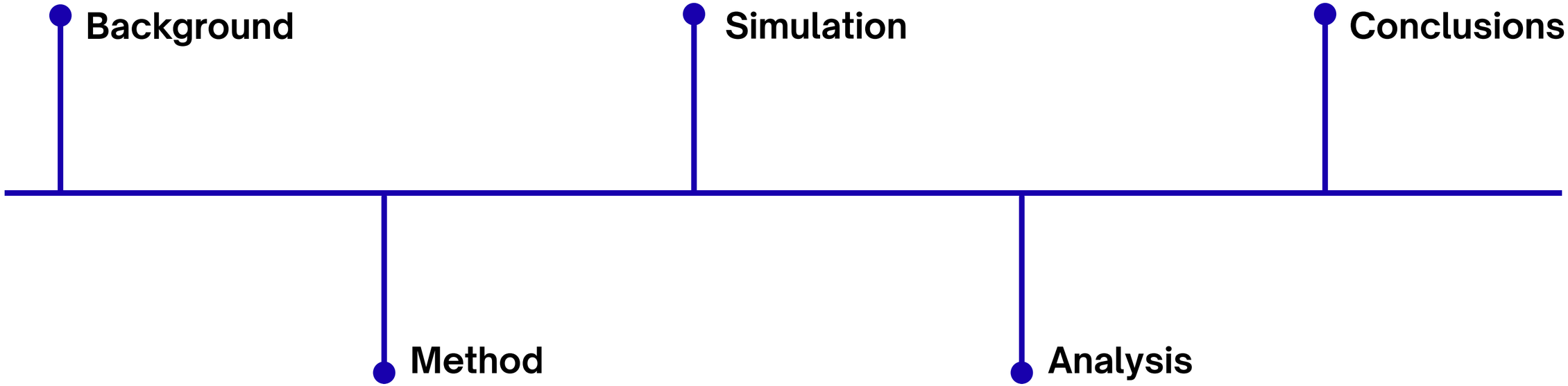


Whole-body Jacobian (WBJ)



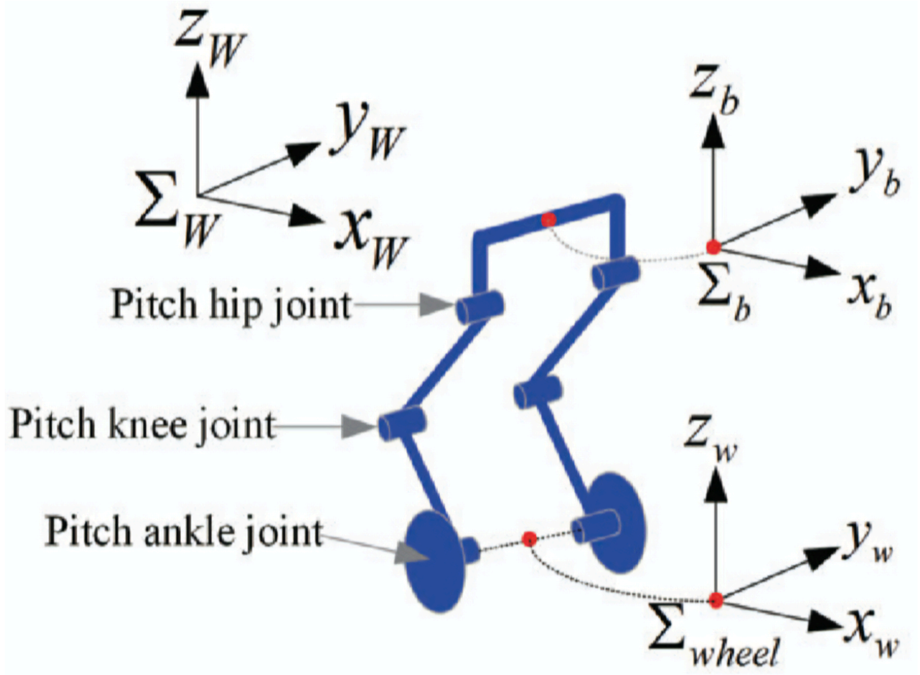
Control via horizontal posture

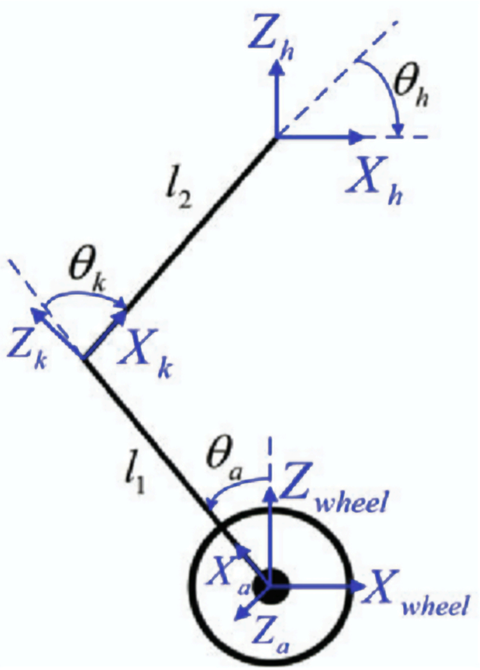
Overview



Background

SINGLE WHEELED-LEG KINEMATICS



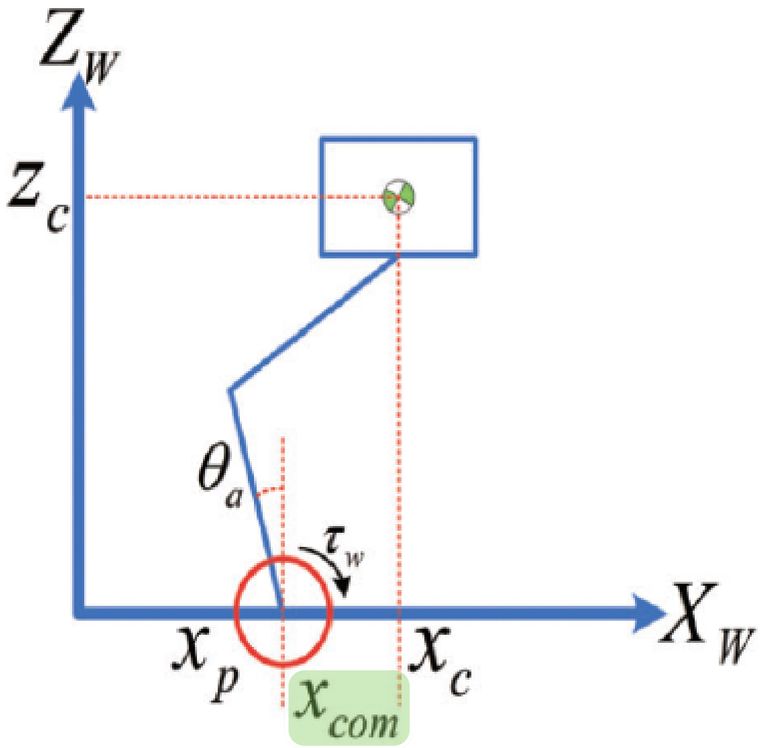


- Σ_W : inertial world frame
- Σ_b : torso frame
- Σ_{wheel} : wheel frame

- θ_a: ankle angle
- θ_k: knee angle
- θ_h: hip angle

Background

INVERTED PENDULUM MODEL



$$\ddot{x}_c = \frac{g}{z_c}(x_c - x_p) = \frac{g}{z_c} x_{com}$$

The CoM acceleration can be modulated by controlling the horizontal posture.

Background

FORWARD KINEMATICS

Joint angles → **Hip pose** → **Task space**

$$\begin{bmatrix} \theta_a^t \\ \theta_k^t \\ \theta_h^t \end{bmatrix} \longrightarrow \begin{bmatrix} x_h^t \\ z_h^t \\ \theta_{\text{pitch}}^t \end{bmatrix} = \begin{bmatrix} -l_1 \sin(\theta_a^t) - l_2 \sin(\theta_a^t + \theta_k^t) \\ l_1 \cos(\theta_a^t) + l_2 \cos(\theta_a^t + \theta_k^t) \\ \theta_a^t + \theta_k^t + \theta_h^t \end{bmatrix} \longrightarrow \begin{bmatrix} x_{com} \\ z_{com} \\ d \\ \beta_{com} \end{bmatrix} = \begin{cases} \dfrac{x_h^l + x_h^r}{2} \\ \dfrac{z_h^l + z_h^r}{2} \\ z_h^r - z_h^l \\ \dfrac{\theta_{\text{pitch}}^l + \theta_{\text{pitch}}^r}{2} \end{cases}$$



- x_h : hip position along the x-axis
- z_h : hip position along the z-axis
- θ_h : orientation of the thigh relative to the wheel frame

- x_{com}, z_{com} : position of the Com
- d : height difference between the two legs
- β_{com} : torso pitch angle

Background

TASK PLANNING

Whole-body jacobian (WBJ) relates the joint velocities of both legs to the time derivative of the torso task variables.

TRANSLATION

$$\dot{q}_{com} = \begin{bmatrix} \dot{x}_{com} \\ 0 \\ \dot{z}_{com} \\ \dot{d} \\ \dot{\beta}_{com} \\ 0 \end{bmatrix} = WBJ \begin{bmatrix} \dot{q}_j^l \\ \dot{q}_j^r \end{bmatrix}$$

- $\dot{\alpha}_{com}$: yaw rate;
- $\hat{\gamma}_{com}$: roll angle;
- v_{com} : foward velocity of the CoM

for additional posture constraints:

$$\theta_{pitch}^l = \theta_{pitch}^r$$

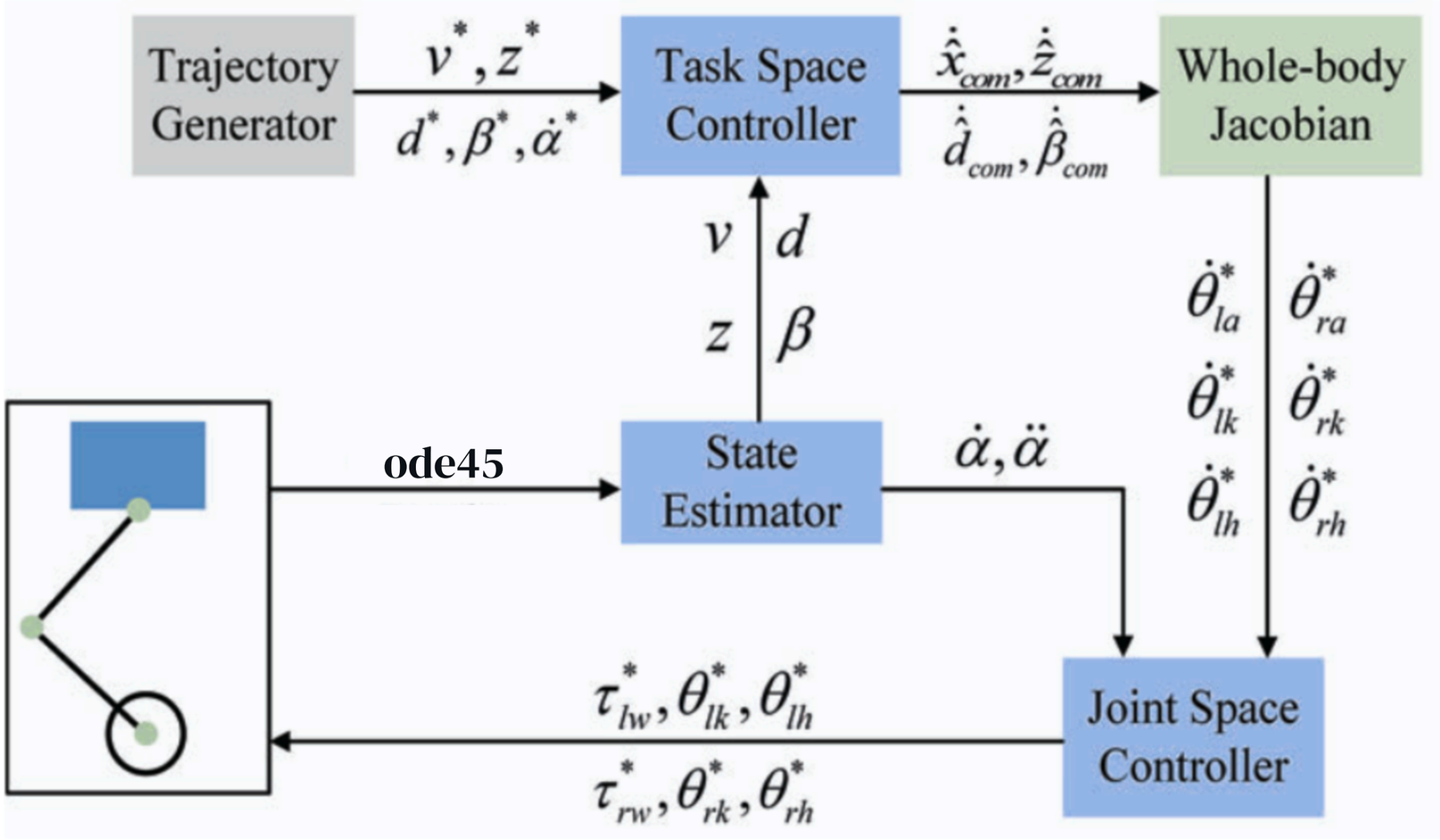
$$x_h^l = x_h^r$$

ROTATION

$$\hat{d} = l_b \tan(\hat{\gamma}_{com}) \qquad \hat{\gamma}_{com} = \arctan\left(\frac{\dot{\alpha}_{com}\, v_{com}}{g}\right)$$

Method

CONTROL ARCHITECTURE



Method

TASK SPACE CONTROLLER

$$\chi = \begin{bmatrix} v_{com} & 0 & z_{com} & d & \beta_{com} & 0 \end{bmatrix}^T \qquad \chi^* = \begin{bmatrix} v^*_{com} & 0 & z^*_{com} & d^* & \beta^*_{com} & 0 \end{bmatrix}^T$$

$$\dot{\hat{q}}_{com} = \begin{bmatrix} \dot{\hat{x}}_{com} \\ 0 \\ \dot{\hat{z}}_{com} \\ \dot{\hat{d}} \\ \dot{\hat{\beta}}_{com} \\ 0 \end{bmatrix} = K_P\left(\chi^* - \chi\right) + K_D\left(\dot{\chi}^* - \dot{\chi}\right)$$

- v_{com} : velocity of the CoM w.r.t. the world frame
- $\dot{x}_{com}$: velocity of the CoM w.r.t the wheel frame

Method

JOINT SPACE CONTROLLER

$$q_j^*(t+1) = q_j^*(t) + \dot{q}_j^* \Delta t$$

Hip and knee

$$\begin{cases} \ddot{q}^t = K_{p,j}(q^{*t} - q^t) + K_{d,j}(\dot{q}^{*t} - \dot{q}^t) \end{cases}$$

Ankle joints are not directly controlled; instead, they are indirectly regulated by applying the torques to the wheels

Ankle joints

$$\begin{cases} \tau_w^t = k_p^t(\theta_a^{t*} - \theta_a^t) + k_d^t(\dot{\theta}_a^{t*} - \dot{\theta}_a^t) \\ \tilde{\tau} = \tilde{k}_p(\dot{\alpha}^* - \dot{\alpha}) + \tilde{k}_d(\ddot{\alpha}^* - \ddot{\alpha}) \end{cases} \quad \begin{cases} \tau_w^{r*} = \tau_w^r - \tilde{\tau} \\ \tau_w^{l*} = \tau_w^l + \tilde{\tau} \end{cases}$$

Simulation

SETTINGS



Direct Drive Tech-Tita Robot

MATLAB

ode45:

$$x_0 = [q^l \quad \dot{q}^l \quad q^r \quad \dot{q}^r \quad v \quad \omega^l \quad \omega^r \quad q^{l*} \quad q^{r*}]^T$$

pipeline:

1) State unpacking
2) Yaw rate computation
3) Reference trajectory evaluation
4) WBJ computation
5) Task space reconstruction
6) Computation of χ and $\dot{\chi}$
7) Task space controller
8) Desired joint velocity computation
9) Wheel torque computation
10) Wheel angular acceleration
11) Joint-space position control
12) State derivative construction

Simulink

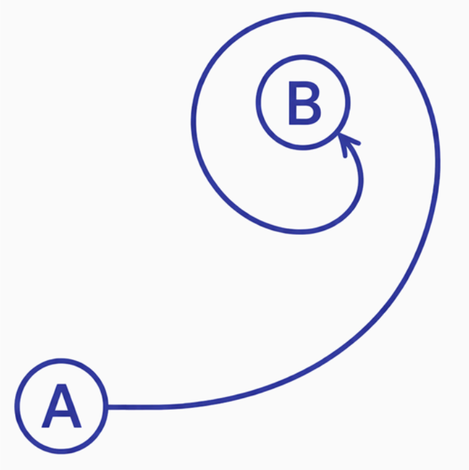
Robot visualization using URDF model and mesh files

Simulation

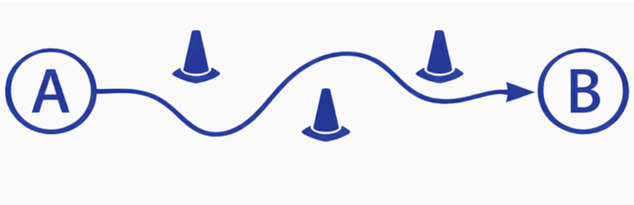
EXPERIMENTS



Velocity Tracking



Velocity and Yaw Tracking



Slalom Trajectory Tracking

Analysis

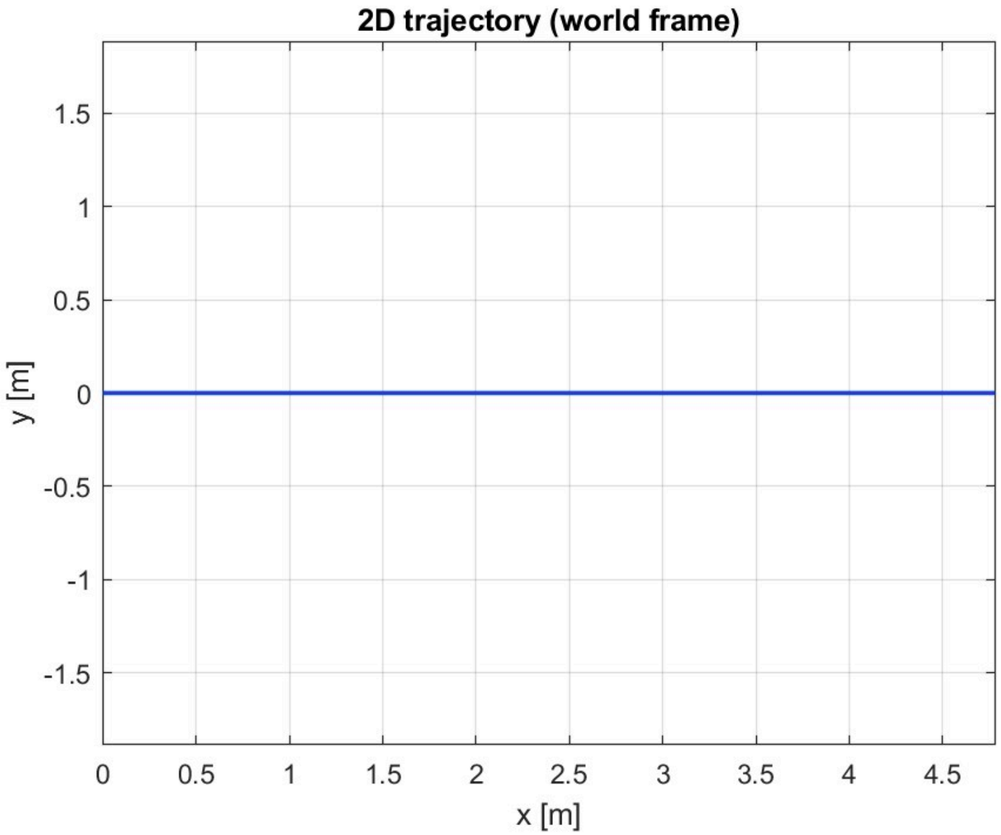
VELOCITY TRACKING TASK

Task definition:

$$\chi^*(t) = [1.5\sin(0.2\pi t) \quad 0 \quad z_0 \quad d_0 \quad \beta_0 \quad 0]^T$$

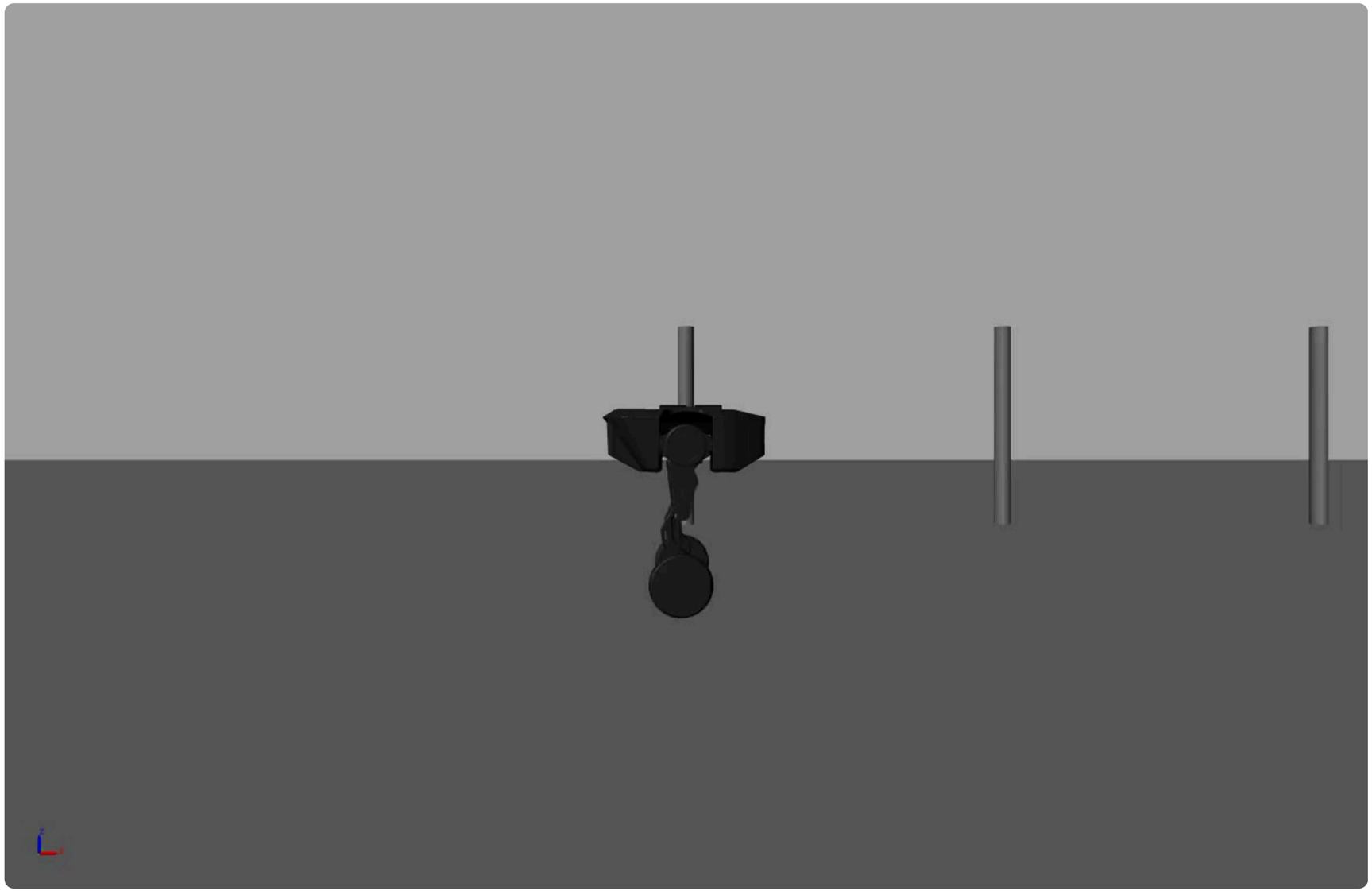
$$\dot{\chi}^*(t) = [0.3\pi\cos(0.2\pi t) \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T$$

$$\dot{\alpha}^*(t) = 0 \qquad \ddot{\alpha}^*(t) = 0$$



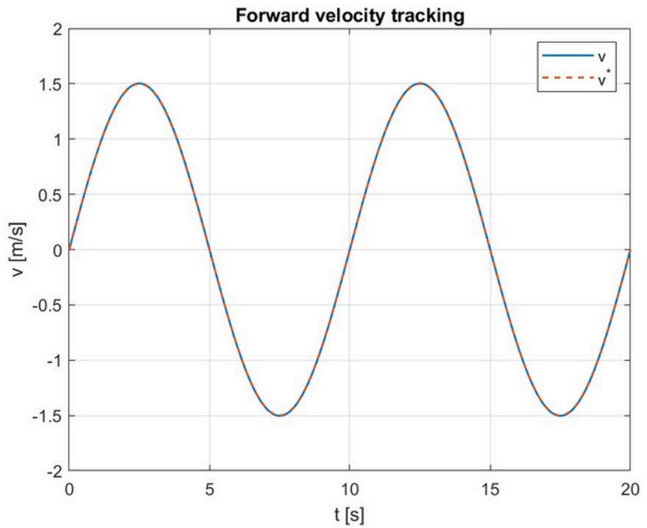
Analysis

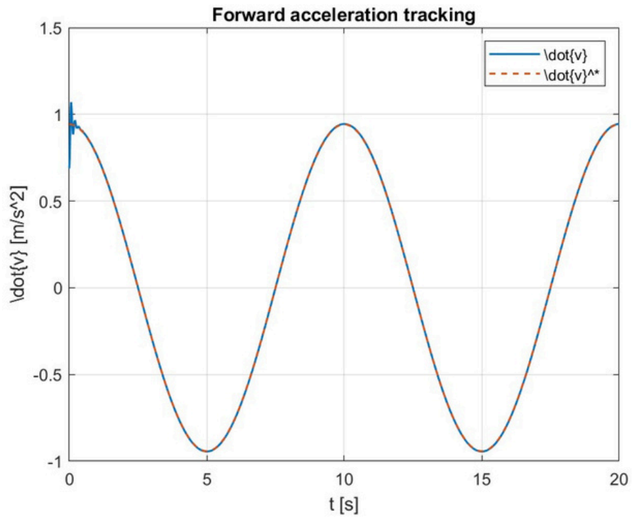
VELOCITY TRACKING TASK

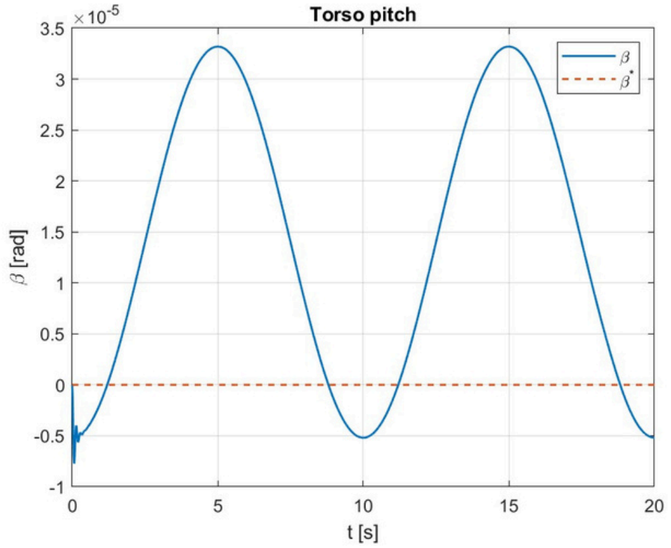


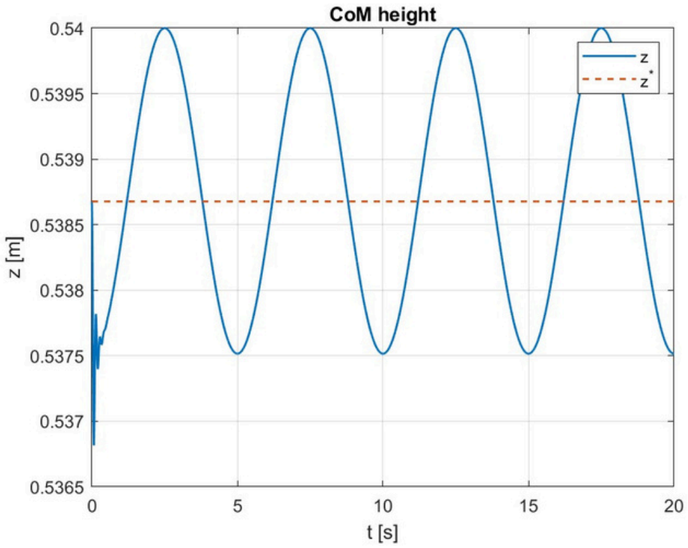
Analysis

VELOCITY TRACKING TASK



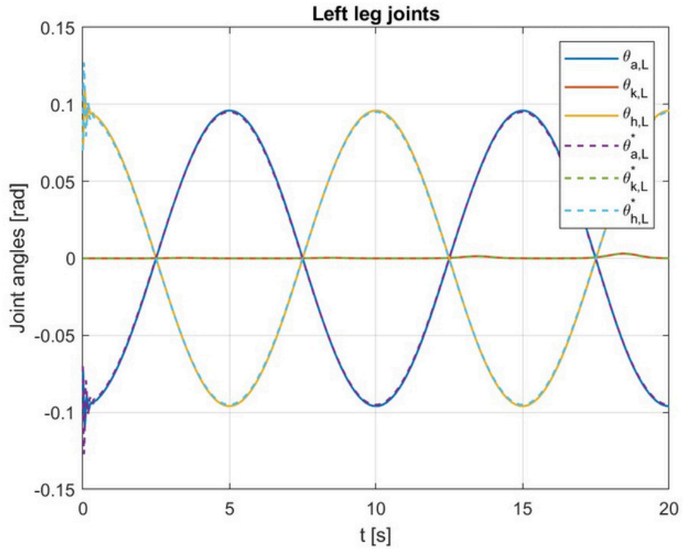


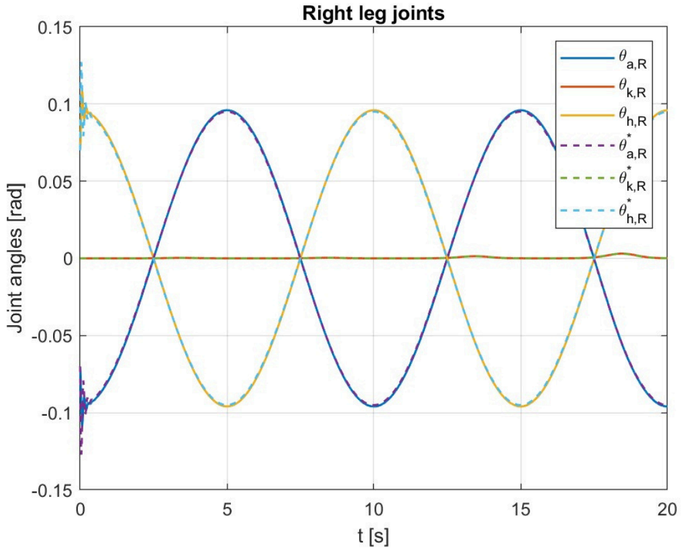


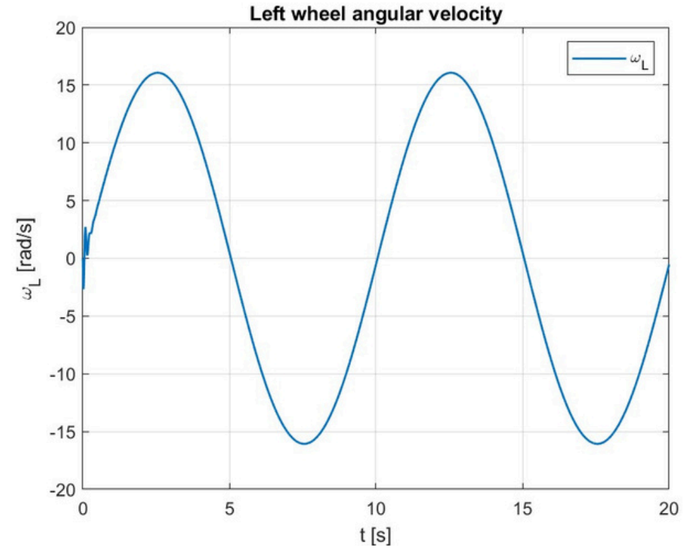


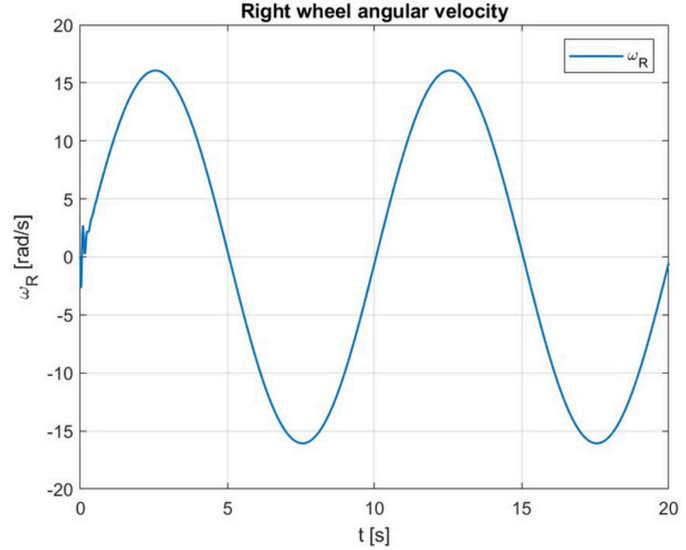
Analysis

VELOCITY TRACKING TASK









Analysis

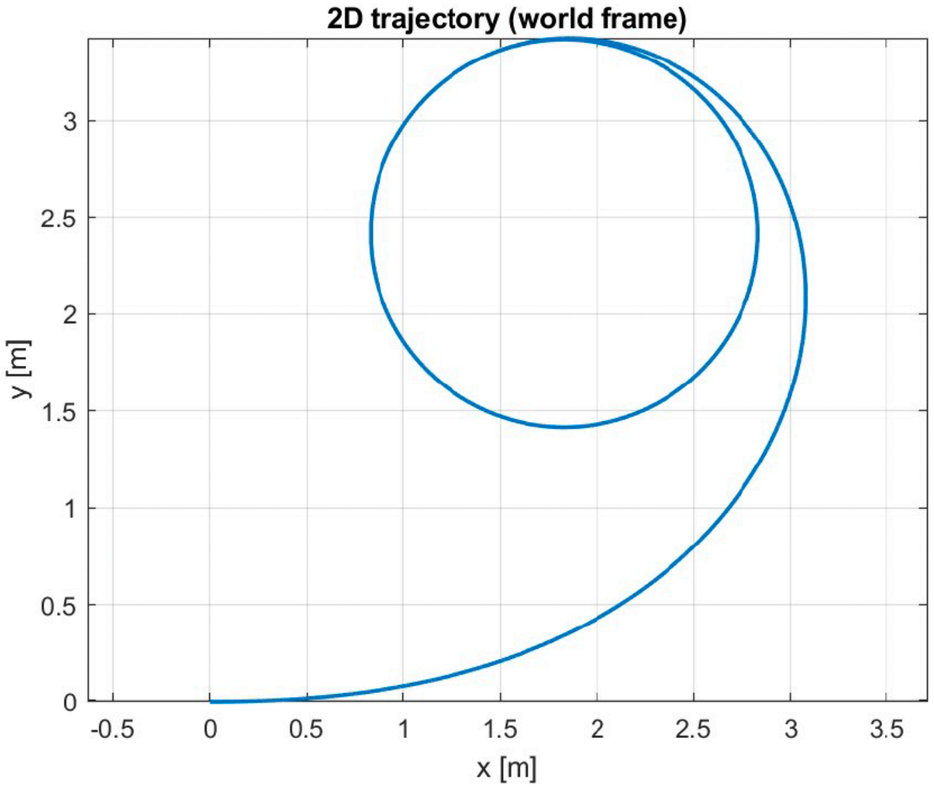
VELOCITY AND YAW TRACKING TASK

Task definition:

$$\dot{\alpha}^*(t) = \begin{cases} 0.3\,t & t \leq 5 \\ 1.5 & t > 5 \end{cases} \quad \ddot{\alpha}^*(t) = \begin{cases} 0.3 & t \leq 5 \\ 0 & t > 5 \end{cases}$$

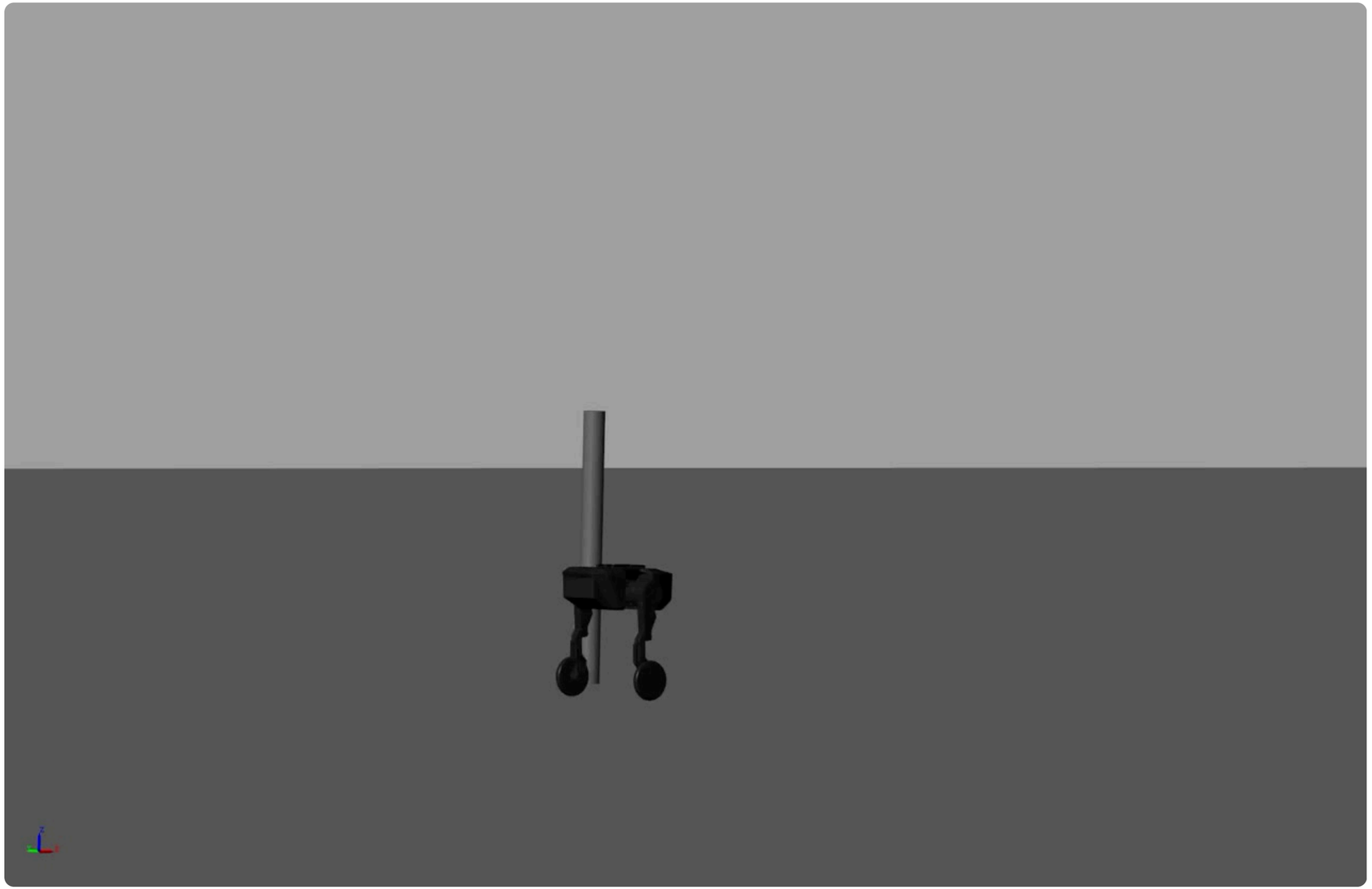
$$\chi^*(t) = \begin{bmatrix} 1.5 & 0 & z_0 & l_b \dfrac{1.5\dot{\alpha}^*(t)}{g} & \beta_0 & 0 \end{bmatrix}^T$$

$$\dot{\chi}^*(t) = \begin{bmatrix} 0 & 0 & 0 & l_b \dfrac{1.5}{g}\ddot{\alpha}^*(t) & 0 & 0 \end{bmatrix}^T$$



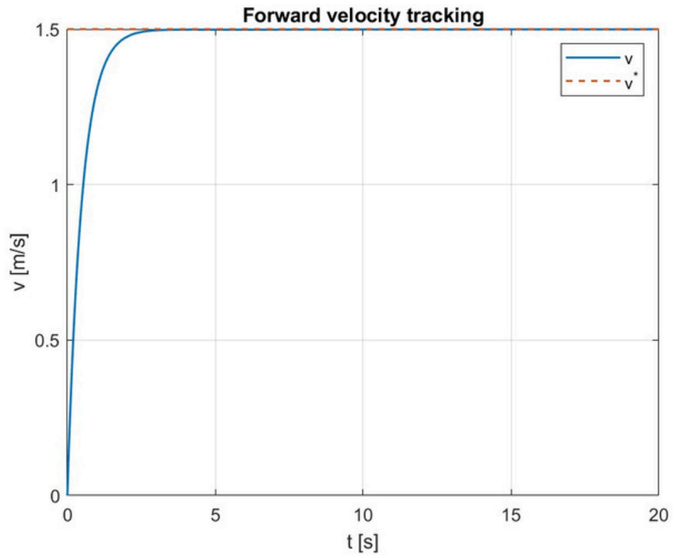
Analysis

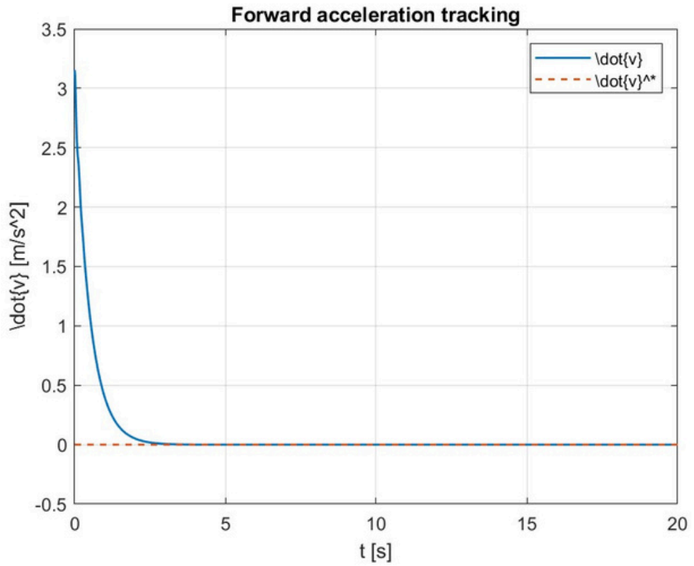
VELOCITY AND YAW TRACKING TASK

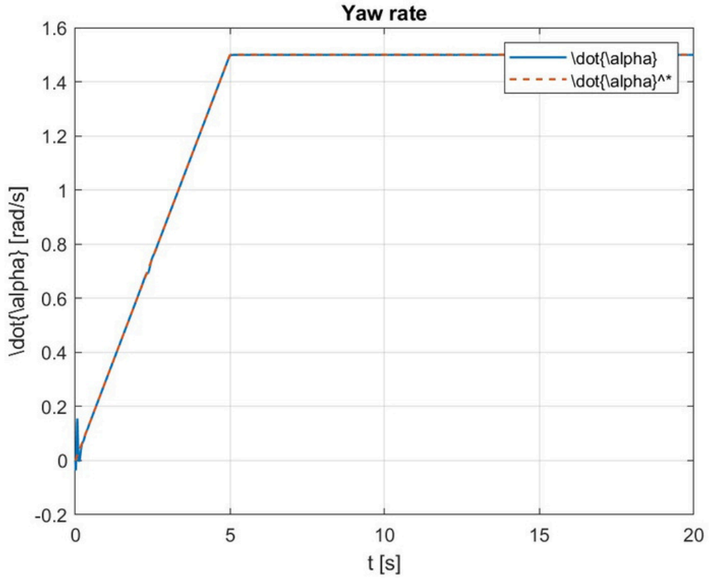


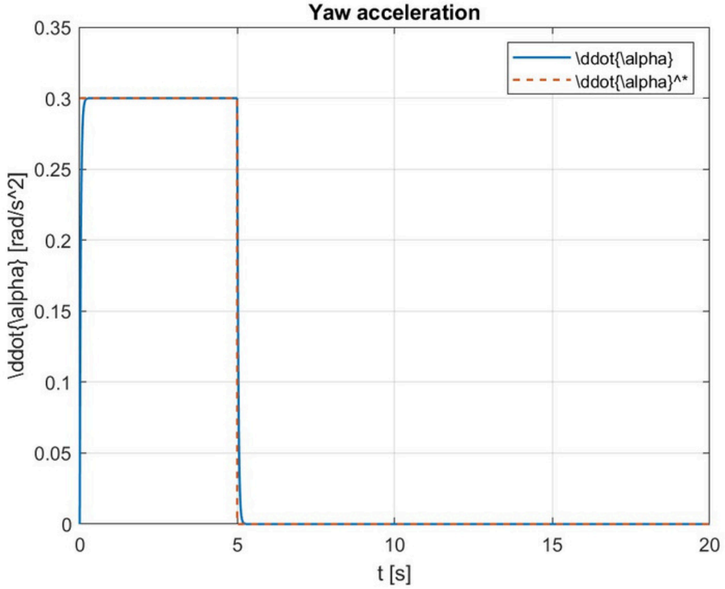
Analysis

VELOCITY AND YAW TRACKING TASK



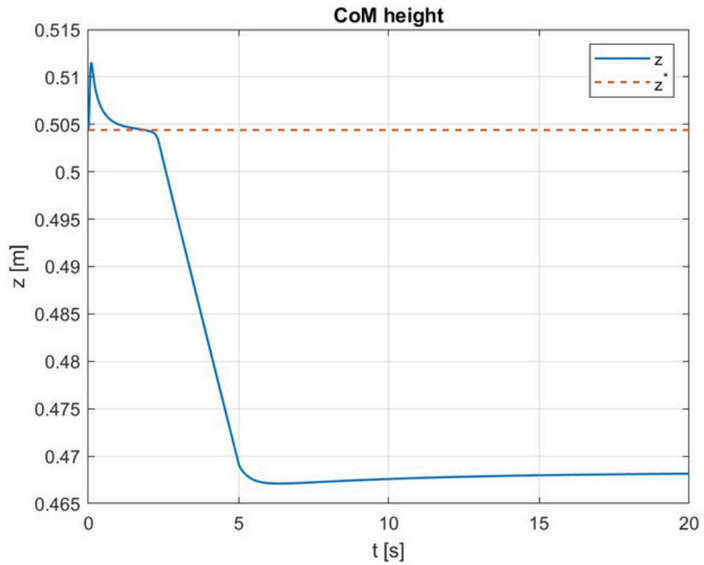




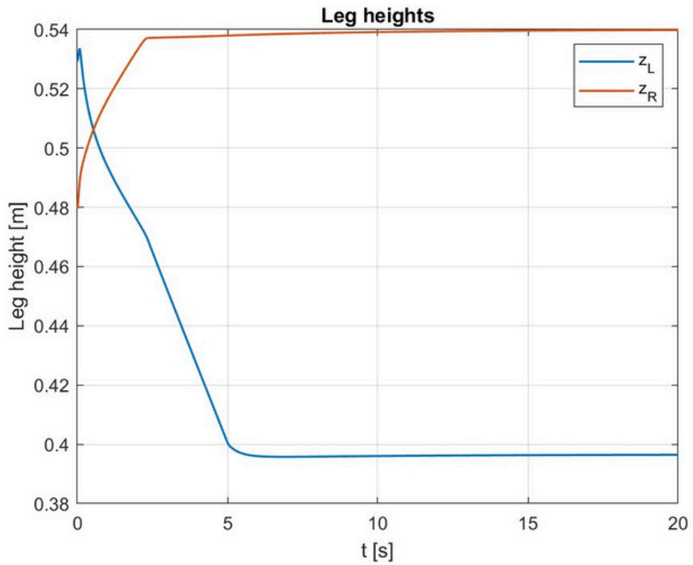


Analysis

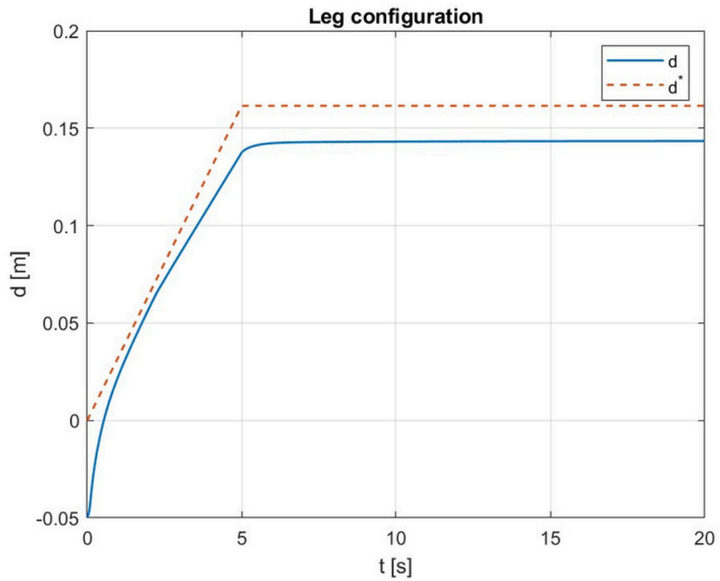
VELOCITY AND YAW TRACKING TASK



CoM height



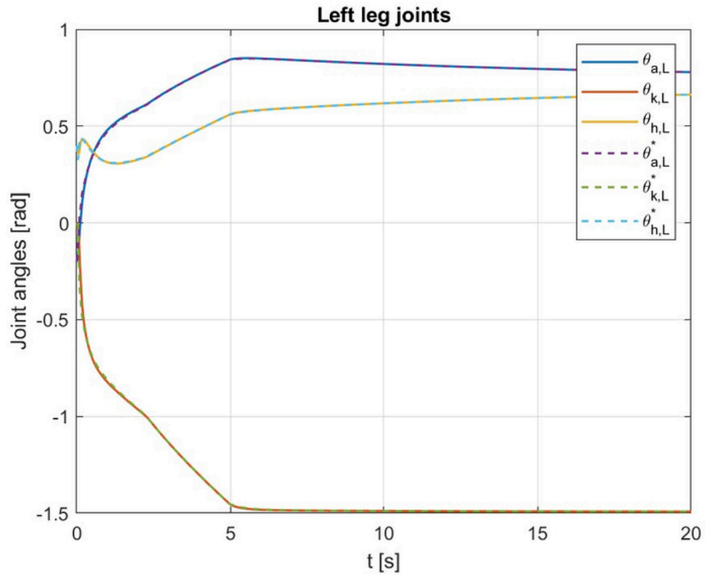
Leg heights

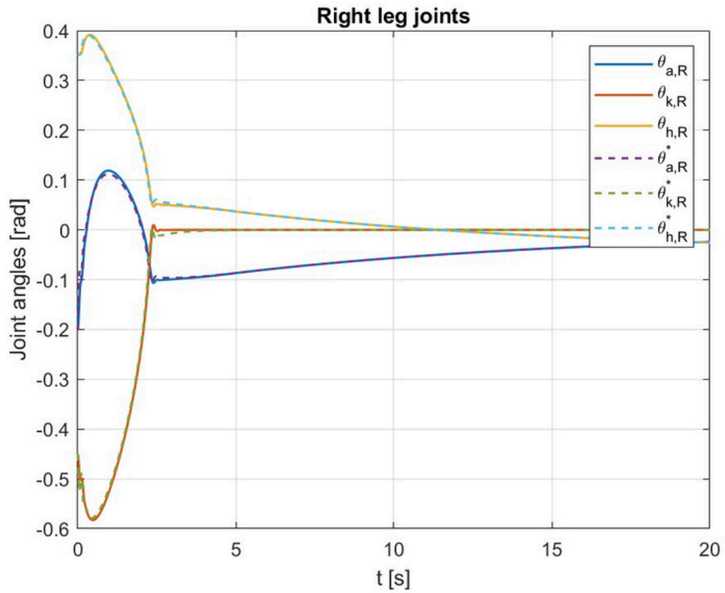


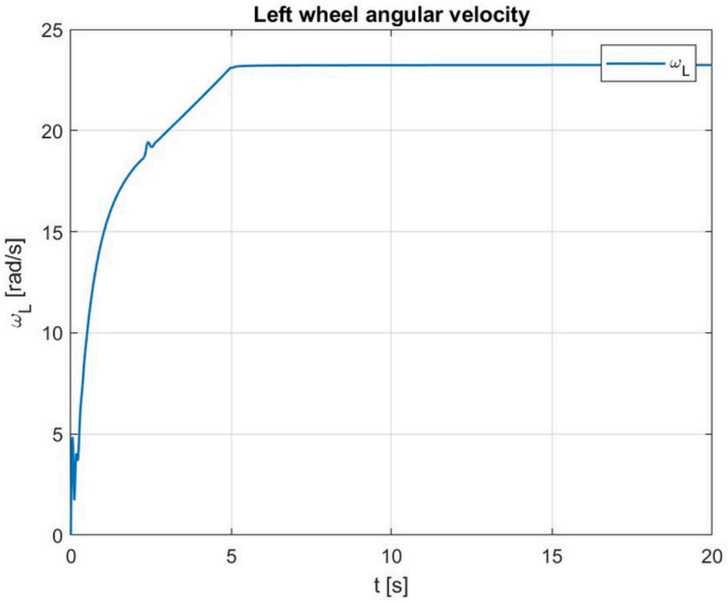
Height difference

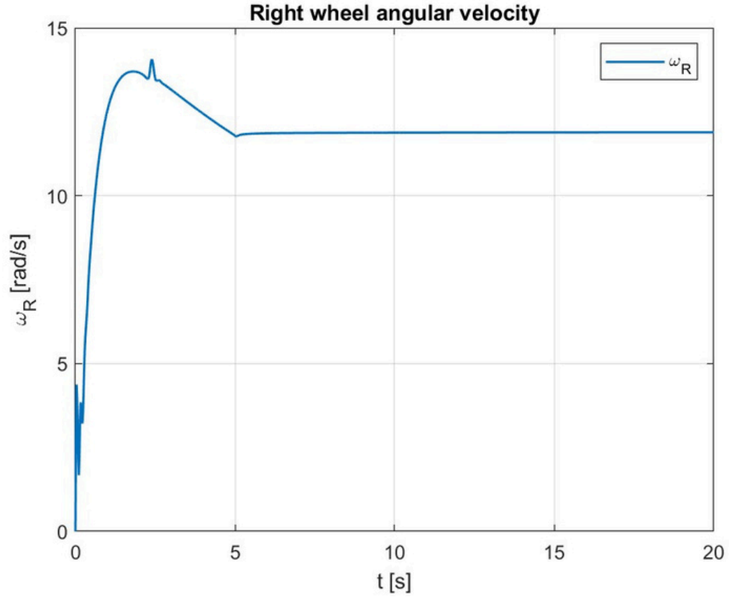
Analysis

VELOCITY AND YAW TRACKING TASK









Analysis

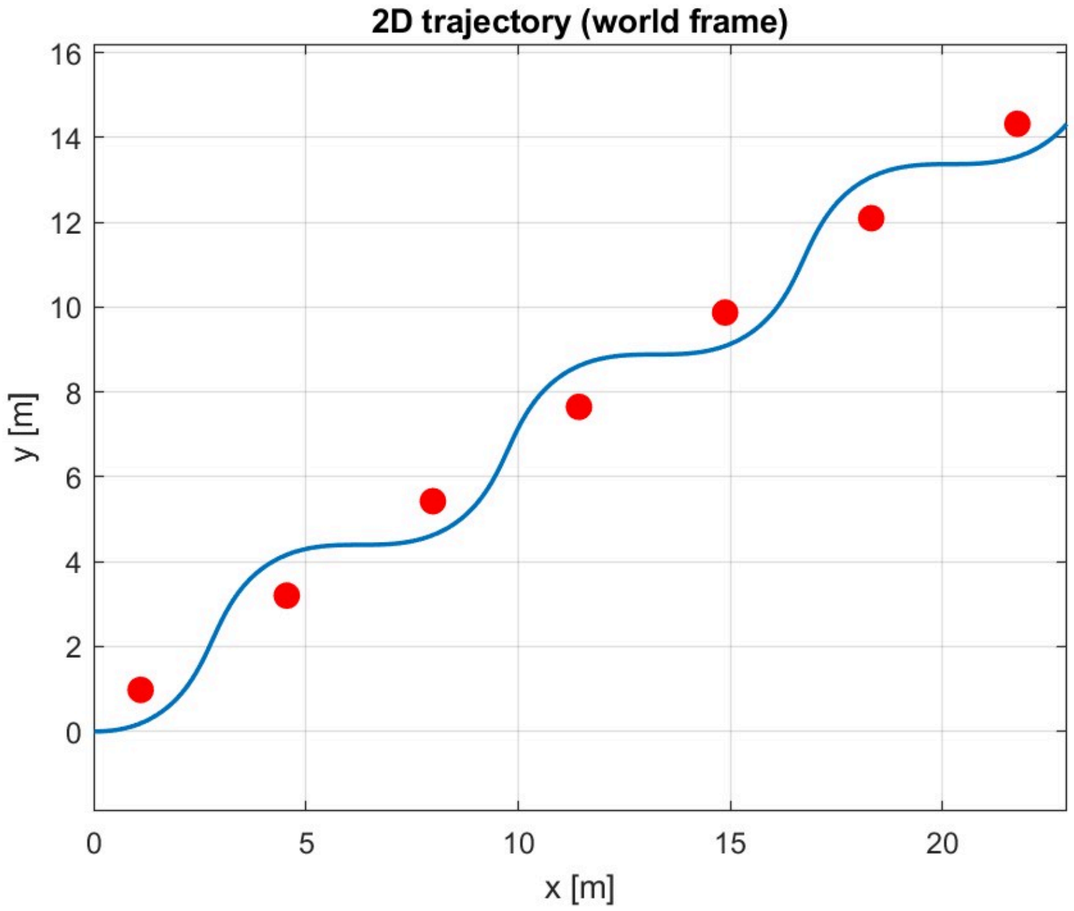
SLALOM TASK

Task definition:

$$\dot{\alpha}^*(t) = A\sin(2\pi f t) \quad \ddot{\alpha}^*(t) = 2\pi f A\cos(2\pi f t)$$

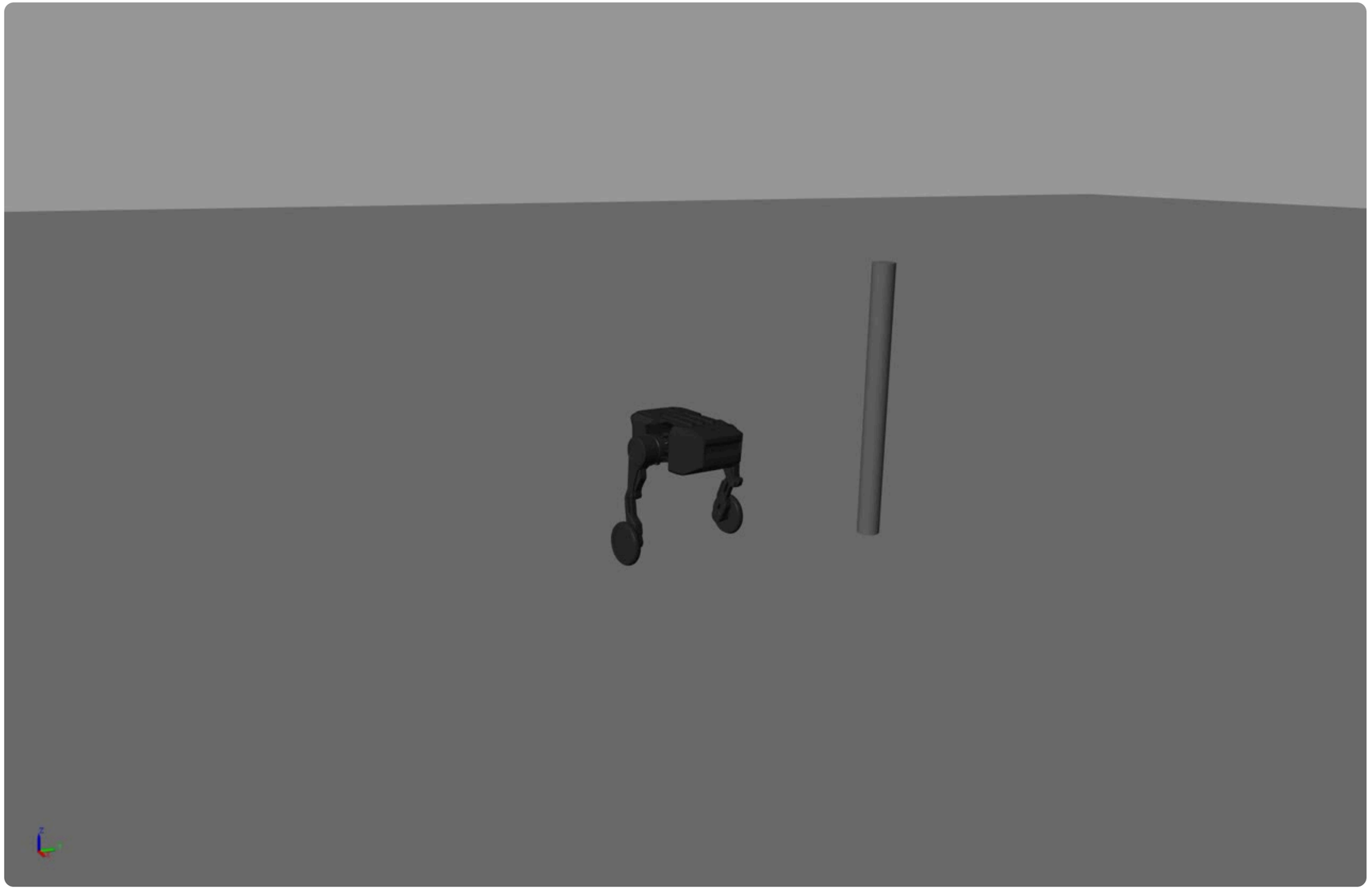
$$\chi^*(t) = \begin{bmatrix} 1.5 & 0 & z_0 & l_b\dfrac{1.5\dot{\alpha}^*(t)}{g} & \beta_0 & 0 \end{bmatrix}^T$$

$$\dot{\chi}^*(t) = \begin{bmatrix} 0 & 0 & 0 & l_b\dfrac{1.5}{g}\ddot{\alpha}^*(t) & 0 & 0 \end{bmatrix}^T$$



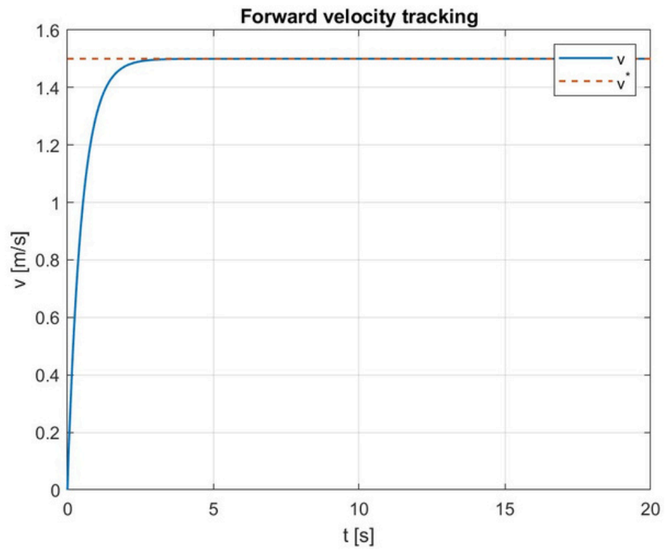
Analysis

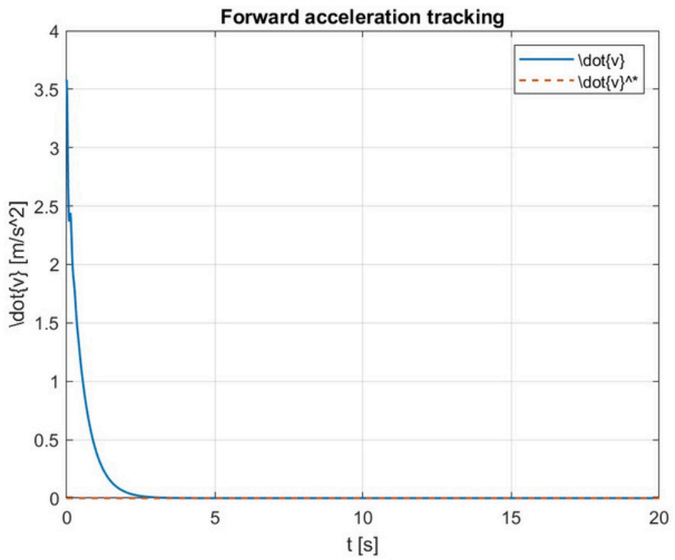
SLALOM TASK

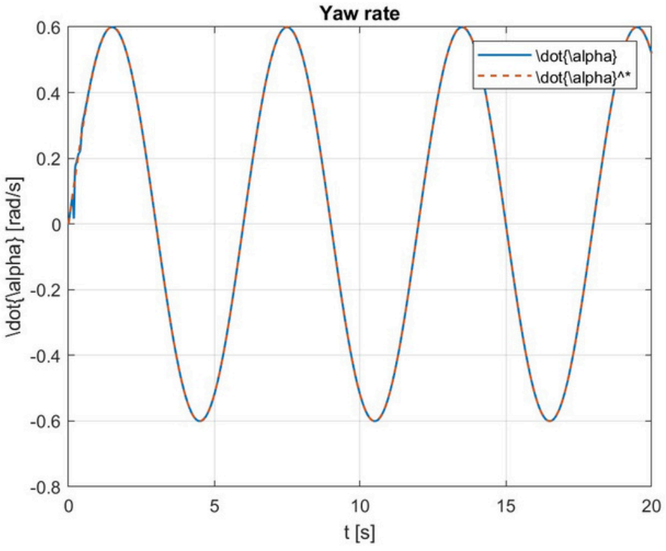


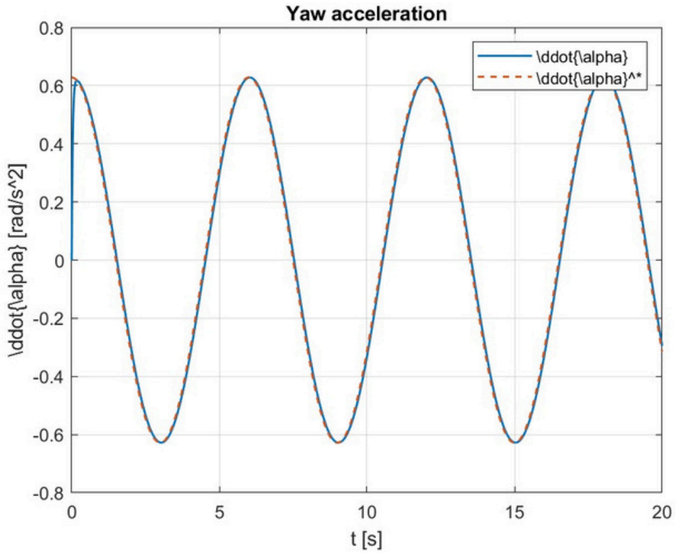
Analysis

SLALOM TASK



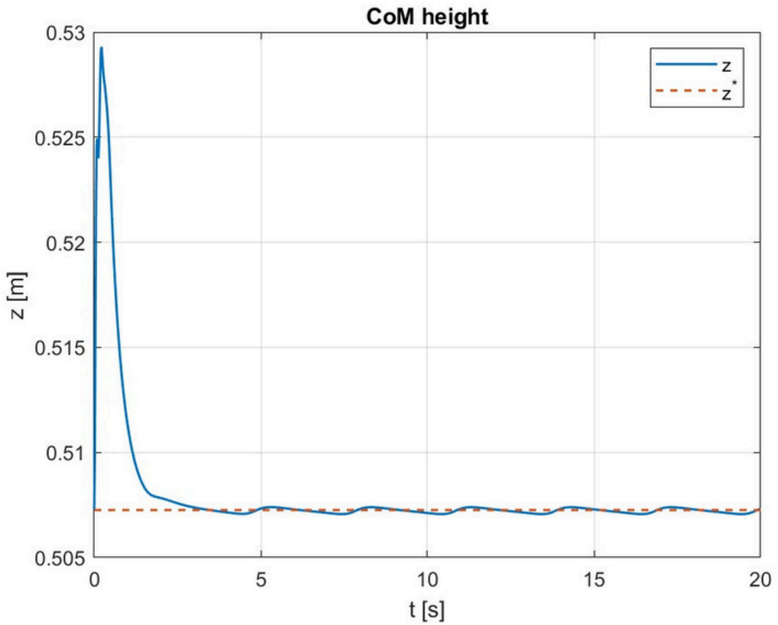




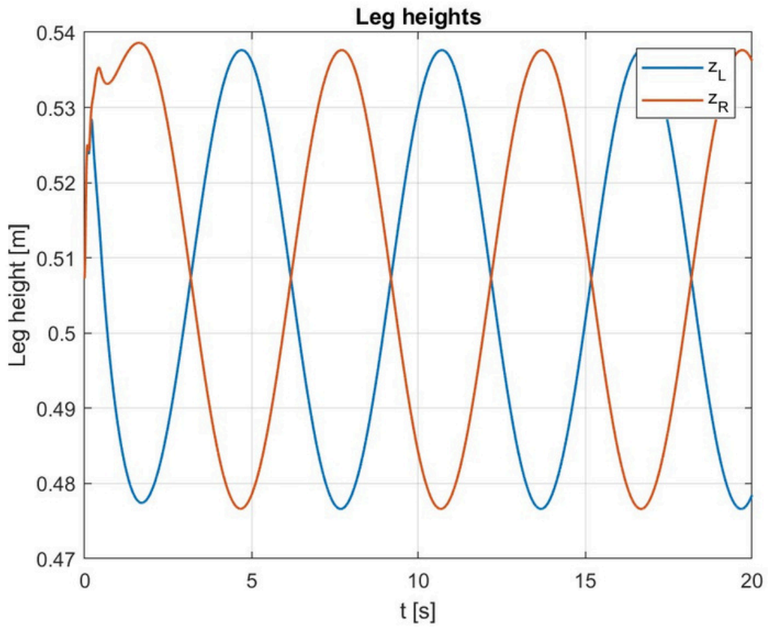


Analysis

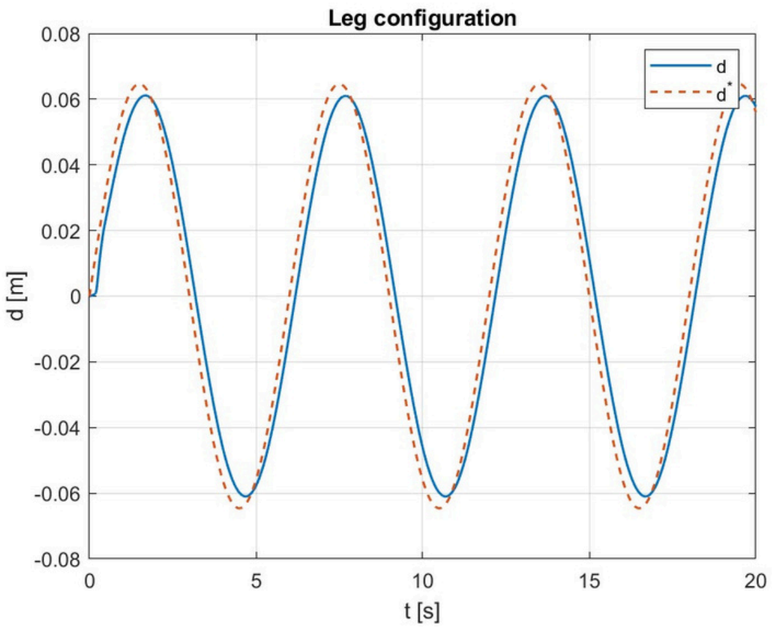
SLALOM TASK



CoM height



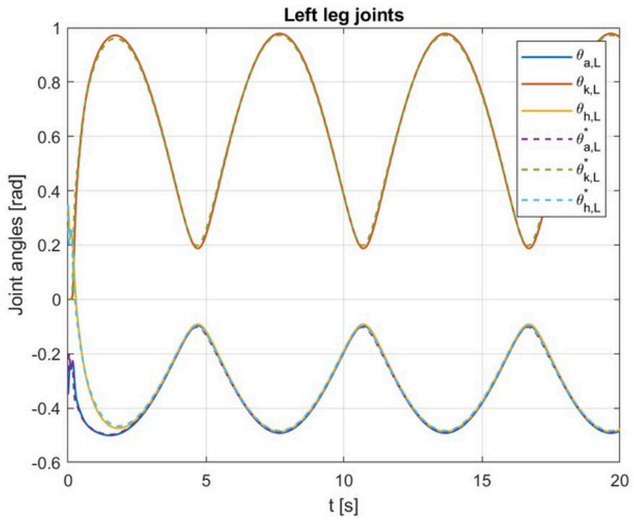
Leg heights

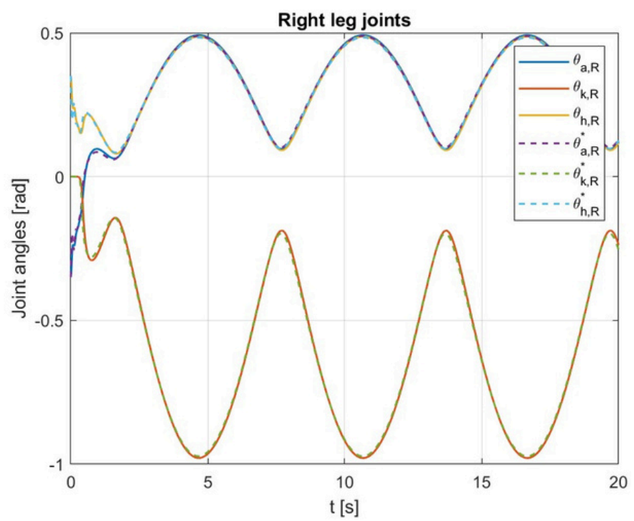


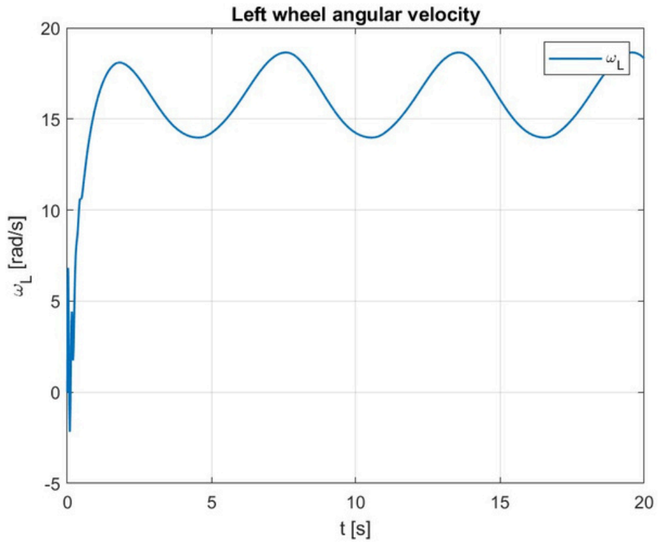
Height difference

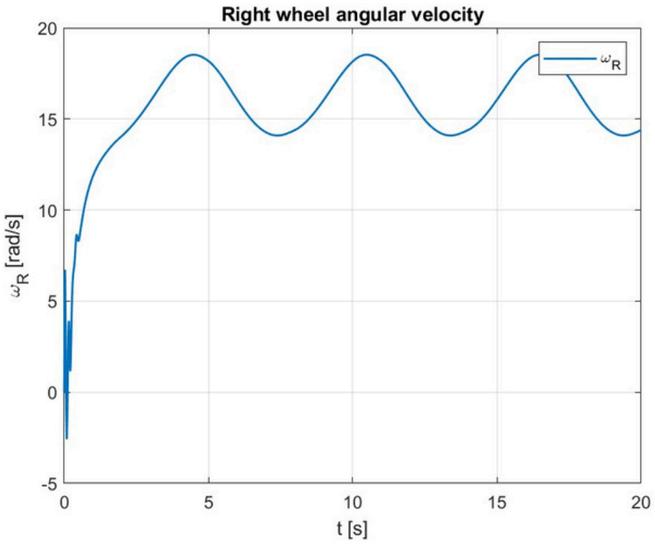
Analysis

SLALOM TASK









Conclusions

- **Whole-body motion planning framework** linking horizontal posture and dynamic balance
- **Whole-Body Jacobian (WBJ)** computation for connection between the task space and the joint space
- **Task space controller** and **joint space controller** for trajectory tracking
- Experiments validate stable and effective motion, including **velocity control** and **turning maneuvers**